PARTH LODAT

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Summary

AI Engineer at Mediallianz Marketing & Digital Media Solutions LLP, building **production LLM systems** end to end — **RAG pipelines**, real-time signal detection and **multi-agent automation**. Architected **PulseLoop**, a four-service AI sales-intelligence platform (FastAPI RAG over pgvector with streaming LLM coaching), an **autonomous 5-agent publishing pipeline** shipping **∼9–10 SEO articles/day** unattended, and a **Google Ads analytics platform** reconciling ad spend against true conversions. **Converted from intern to permanent AI Engineer in 6 months.**

Skills

Frontend Development: React.js, Next.js, JavaScript (ES6+), HTML5, CSS3

AI / ML: Machine Learning, NLP, Recommendation Systems, RAG Pipeline, Embeddings, LLM Agents

Frameworks / Libraries: LangChain, Pytorch, Scikit-learn and Hugging Face Transformers

AI Automation: n8n, Make.com

Programming / Database: Python, JavaScript, SQL, PostgreSQL

Experience

Mediallianz Marketing & Digital Media Solutions LLP — **Mumbai, India**

AI Engineer (promoted from AI Engineer Intern, Jan–Jul 2026) — *Jan 2026 – Present*

PulseLoop — AI Sales Intelligence Platform (RAG)

- Architected a **4-service platform** (FastAPI RAG, Node/TypeScript API, background worker, React SPA) on **Postgres + pgvector, Redis and Docker Compose** across **14 versioned migrations**.
- Built the **RAG engine** end to end — PDF ingestion with **PyMuPDF + OCR fallback**, chunking, **NVIDIA NIM embeddings** into pgvector and **SSE token-streamed** chat, pitch and creative generation.
- Engineered **real-time call signal detection** over **45-second transcript windows**: a rule-based gate feeds a **difficulty filter** escalating only ambiguous windows to the LLM — **cutting LLM calls to the hard cases**.
- Shipped a **post-call coaching generator** using **hierarchical retrieval** (account + global tiers), plus **role-based dashboards** for Rep, Manager, L&D and Admin with JWT auth.

Real Doodle — Autonomous AI Publishing Pipeline (realdoodle.com)

- Built a **5-agent LLM pipeline** (Research → Writer → SEO → Quality → Publish) in **n8n + GPT** running every 2.5 hours — **∼9–10 formatted 850–1100 word articles/day (∼290/month)**, zero manual editing.
- Grounded every article in **live search results** with fact-only prompts (**eliminating hallucinated scores, quotes and statistics**) and engineered a **story-key dedup algorithm** across **60+ national teams**.
- Automated the publishing tail — image sourcing, WordPress upload, SEO meta, categories, tagging — at **1 execution per article**, fitting a 2,500 exec/month limit.

Google Ads Analytics & CPA Reconciliation Dashboard

- Built a **1,200+ line Next.js / React dashboard** (Recharts, dark mode, PNG/CSV export) with date-range filtering and **6 comparison modes**.
- Designed an **n8n pipeline** pulling **Google Ads API v23** metrics and parsing daily GCLID reports out of Gmail into one source of truth — **replacing a manual daily spreadsheet reconciliation**.
- Implemented the **reported-vs-actual reconciliation** the business runs on — Actual CPA, Dash CPA, Diff %, A-CR, D-CR per campaign across **3 markets** — with **incremental sync** to cut redundant API calls.

Lead Qualification & Enrichment Automation

- Automated **Facebook Lead Ads → enrichment → CRM → routed follow-up** in Make.com, scoring leads Qualified / Warm / Disqualified and firing **instant Twilio SMS and email** — **response time from hours to seconds**.

Projects

1) Financial Chatbot

- Developed a **financial domain chatbot** using **RAG** for accurate, context-aware answers, with an end-to-end pipeline for **PDF ingestion → text chunking → embeddings → semantic retrieval** on **Chroma DB**.
- Built frontend UI using **Next.js + React** with reusable component structure, integrated chatbot APIs using **Next.js API routes** and implemented secure login via **NextAuth**.

2) Movie Recommendation System

- Developed a **Movie Recommendation System** that provided personalized recommendations using **similarity-based Machine Learning**.
- Built an interactive **Streamlit** UI with a **Trending Movies carousel**, integrated **TMDB API** for posters and metadata, and used **serialized artifacts** for fast inference.

Education

B.E. Computer Engineering — **2022 – Present**